

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Comparative Analysis

Code of Conduct:

I PAVAN JANA (192472313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: PAVAN j

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	

Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5

B 3.0 3.8

C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

2. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

3. Precision-Recall Trade-off Analysis

Two models produce:

Model TP FP FN

A 70 30 20

B 85 25 25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

4. F1-Score Comparison

Two models:

Model Precision Recall

A 0.8 0.6

B 0.75 0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

5. Optical Flow Consistency

Motion values:

Frame Method A Method B

1 4 6

2 5 8

3 6 9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

1. Depth Estimation Comparison — Stereo Vision vs Structure from Motion (SfM)

Introduction

Depth estimation is an important task in computer vision. It determines the distance between a camera and objects in a scene. Two common techniques are **Stereo Vision** and **Structure from Motion (SfM)**.

- **Stereo Vision** estimates depth using two or more images captured from different camera positions.
- **Structure from Motion (SfM)** estimates 3D structure and camera motion from multiple images taken from different viewpoints.

The accuracy of these techniques can be evaluated by comparing their estimated depth values with the actual depth values.

Given Data

Object	Actual Depth (m)	Stereo Vision (m)	SfM (m)
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A	2.0	2.0	2.5
B	3.0	3.0	3.8
C	4.0	4.0	5.0

(a) Calculate Absolute Error

Absolute error tells us **how far the estimated value is from the actual value**, without considering whether the error is positive or negative.

Formula

A smaller absolute error means the method is more accurate.

Stereo Vision

Object A

Actual	depth	=	2.0	m
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Estimated depth = 2.0 m

Object B

Object C

Therefore:

Mean Absolute Error

Mean Absolute Error (MAE) gives the average error across all objects.

SfM

Object A

Object B

Object C

Therefore:

Mean Absolute Error

Detailed Comparison

	Object	Actual	Stereo Estimate	Stereo Error	SfM Estimate	SfM Error
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A	2.0	2.0	0.0	2.5	0.5
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B	3.0	3.0	0.0	3.8	0.8
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C	4.0	4.0	0.0	5.0	1.0
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MAE			0.000		0.767
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Additional Observation

SfM overestimates the depth for every object:

- Object A is overestimated by **0.5 m**
- Object B is overestimated by **0.8 m**
- Object C is overestimated by **1.0 m**

The error also increases as the actual depth increases.

Evaluation

Stereo Vision has:

SfM has:

Since **lower error means better accuracy**, Stereo Vision clearly performs better.

Final Conclusion

Stereo Vision is the better technique for this dataset. It produces exactly the actual depth values

for all three objects, resulting in zero absolute error and zero MAE. SfM produces larger depth estimates than the actual values and has an average error of approximately **0.767 m**.

2. Pattern Recognition Accuracy Comparison

Introduction

Pattern recognition is the process of identifying patterns or categories from input data. For example, an image recognition system may classify an image as a cat, dog, vehicle, or person.

The most basic performance measure for a classification system is **accuracy**.

Accuracy tells us the percentage of predictions that are correct among all predictions.

Given Data

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Calculate Accuracy

Formula

Traditional System

Correct predictions = 85

Total predictions = 100

This means the traditional system correctly recognizes **85 out of every 100 samples**.

Incorrect predictions:

So, the traditional system makes **15 incorrect predictions**.

Deep Learning System

Correct predictions = 92

Total predictions = 100

This means the deep learning system correctly recognizes **92 out of every 100 samples**.

Incorrect predictions:

So, the deep learning system makes only **8 incorrect predictions**.

Comparison

Performance Measure Traditional Deep Learning

Correct Predictions	85	92
Incorrect Predictions	15	8
Total Samples	100	100
Accuracy	85%	92%

Accuracy Improvement

Therefore, the deep learning model has a **7 percentage-point improvement in accuracy**.

If we express the improvement relative to the traditional system:

So, the deep learning system's accuracy is approximately **8.24% higher relative to the traditional system**.

Evaluation

The deep learning system makes:
fewer mistakes per 100 samples.

Conclusion

The **Deep Learning system performs better** because it achieves **92% accuracy**, while the traditional system achieves only **85%**.

The deep learning system correctly recognizes 92 samples out of 100 and makes only 8 errors. Therefore, based on accuracy, deep learning is the preferred approach.

Note: Accuracy alone may not be sufficient when classes are highly imbalanced. In such cases, precision, recall, and F1-score should also be considered.

3. Precision-Recall Trade-off Analysis

Introduction

In classification problems, accuracy does not always provide enough information. Two important measures are **precision** and **recall**.

Precision

Precision measures how reliable the positive predictions are.

In simple words:

Out of everything the model predicted as positive, how many were actually positive?

Recall

Recall measures how many actual positive cases were successfully detected.

In simple words:

Out of all actual positive cases, how many did the model find?

Given Data

Model	TP	FP	FN
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A	70	30	20
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B	85	25	25
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Where:

- **TP (True Positive):** Correctly identified positive cases.
- **FP (False Positive):** Negative cases incorrectly identified as positive.
- **FN (False Negative):** Positive cases incorrectly identified as negative.

Model A

Precision

Formula:

Substitute the values:

Therefore:

Interpretation

When Model A predicts a sample as positive, it is correct approximately **70% of the time**.

Recall

Formula:

Substitute:

Therefore:

Interpretation

Model A successfully detects approximately **77.78% of all actual positive cases**.

Model B

Precision

Therefore:

Recall

Therefore:

Comparison

Measure	Model A	Model B
True Positive	70	85
False Positive	30	25
False Negative	20	25
Precision	70%	77.27%
Recall	77.78%	77.27%

Precision Comparison

Model B has higher precision:

Therefore, Model B makes more reliable positive predictions.

Recall Comparison

Model A has slightly higher recall:

Therefore, Model A detects slightly more of the actual positive cases.

Balance Analysis

A balanced model should have precision and recall values that are close to each other.

Model A

Model B

Therefore, Model B is perfectly balanced between precision and recall for these values.

Conclusion

Model B is the more balanced model.

Model A has slightly better recall, but its precision is considerably lower. Model B has nearly equal precision and recall, both around **77.27%**.

Therefore, if the goal is to achieve a good balance between correctly identifying positive cases and avoiding false positive predictions, **Model B is preferable**.

4. F1-Score Comparison

Introduction

F1-score is an important classification performance measure that combines **precision and recall into a single value**.

It is particularly useful when we want a balance between precision and recall.

The F1-score ranges from:

A value closer to **1** indicates better performance.

Given Data

Model	Precision	Recall
A	0.80	0.60
B	0.75	0.75

Formula

Model A

Precision:

Recall:

Substitute:

First calculate the numerator:

Then:

Therefore:

or approximately:

Model B

Precision:

Recall:

Therefore:

Comparison

Model Precision Recall F1-Score

A	0.80	0.60	0.686
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B	0.75	0.75	0.750
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Interpretation

Model A has better precision:

However, its recall is much lower:

Model B has equal precision and recall:

This balance gives Model B a higher F1-score.

Difference

Therefore, Model B's F1-score is approximately **0.064 higher**.

Conclusion

Model B is better overall when both precision and recall are important.

Model A may be useful when avoiding false positives is the main priority because its precision is 80%. However, Model B detects more actual positive cases and maintains a better balance between precision and recall.

Therefore:

5. Optical Flow Consistency

Introduction

Optical flow is a computer vision technique used to estimate the apparent movement of objects or pixels between consecutive video frames.

It is commonly used in:

- Motion detection
- Object tracking
- Autonomous vehicles
- Video surveillance
- Action recognition
- Robot navigation
- Traffic monitoring

Here, two methods are being compared based on their estimated motion values over three frames.

Given Data

Frame Method A Method B

1	4	6
2	5	8
3	6	9

(a) Calculate Average Motion

Average motion tells us the typical amount of motion detected by a method across all frames.

Formula

Method A

Motion values:

Sum:

Average:

Method B

Motion values:

Sum:

Average:

Average Motion Comparison

Method Frame 1 Frame 2 Frame 3 Average

A	4	5	6	5.00
B	6	8	9	7.67

Method B detects a higher average motion than Method A.

However, **higher average motion does not automatically mean better performance**. To determine stability, we need to examine the variation in motion values.

(b) Consistency Analysis

A stable optical flow method should produce relatively consistent motion estimates across frames.

One simple measure of consistency is **range**.

Range Formula

Method A

$$\begin{aligned} \text{Maximum} &= 6 \\ \text{Minimum} &= 4 \end{aligned}$$

Method B

$$\begin{aligned} \text{Maximum} &= 9 \\ \text{Minimum} &= 6 \end{aligned}$$

Since Method A has a smaller range, its motion values fluctuate less.

Standard Deviation Analysis

Standard deviation provides a more detailed measure of variation.

A lower standard deviation indicates that the values are closer to their average and therefore more consistent.

For Method A:

Average:

Values:

Differences from mean:

Squared differences:

Variance:

Standard deviation:

Method B

Average:

The standard deviation is approximately:

Therefore:

So Method A is more consistent.

Final Comparison

Performance Measure	Method A	Method B
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Average Motion	5.00	7.67
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Minimum Motion	4	6
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Maximum Motion	6	9
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Range	2	3
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Standard Deviation	0.82	1.25
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Stability	Higher	Lower
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Important Interpretation

Method B detects a greater amount of average motion:

But its motion values vary more:

Therefore, Method B is more sensitive to motion, but Method A is more stable.

Conclusion

If the requirement is **stable and consistent motion estimation**, Method A is the better choice.

If the requirement is specifically to detect **larger motion values**, Method B may be useful.

However, based purely on consistency and stability, **Method A performs better**.

Overall Performance Evaluation

Question	Measure Used	Technique 1	Technique 2	Best
1. Depth Estimation	MAE	Stereo = 0 m	SfM = 0.767 m	Stereo Vision

Question	Measure Used	Technique 1	Technique 2	Best
2. Pattern Recognition	Accuracy	85%	92%	Deep Learning
3. Precision-Recall	Precision & Recall	70%, 77.78%	77.27%, 77.27%	Model B
4. F1-Score	F1	0.686	0.750	Model B
5. Optical Flow	Average & SD	Avg=5, SD=0.82	Avg=7.67, SD=1.25	Method A for stability

Final Evaluation and Justification

The calculated performance measures show that different techniques are better for different objectives.

For depth estimation, Stereo Vision is clearly superior because it has an MAE of **0 m**, meaning that all its estimated depths exactly match the actual depths. SfM has an MAE of **0.767 m**, so its estimates are less accurate for the given data.

For pattern recognition, Deep Learning performs better because its accuracy is **92%**, compared with **85%** for the traditional system. It also produces fewer incorrect predictions, making it the better choice based on accuracy.

For precision and recall, Model B is more balanced. Model A has slightly better recall (**77.78%**) but lower precision (**70%**). Model B has both precision and recall at approximately **77.27%**, giving it a much better balance.

For F1-score, Model B again performs better. Its F1-score is **0.75**, while Model A achieves approximately **0.686**. Since F1-score combines precision and recall, Model B provides better overall classification performance when both measures are important.

For optical flow, Method B has a higher average motion value (**7.67**) than Method A (**5.00**). However, Method A has a smaller range and lower standard deviation (**0.82** compared with **1.25**), meaning that its motion estimates fluctuate less. Therefore, Method A is considered more stable and consistent.

Final Conclusion

Thus, the **best technique depends on the performance requirement**. Error-based measures favor methods with lower error, accuracy favors higher correct predictions, precision-recall analysis

evaluates classification balance, F1-score evaluates the combined classification performance, and standard deviation helps determine the consistency of motion estimation.